

Audio Amplifier

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Data Sheet specifications

MAXIMUM RATINGS

Rating	Symbol	BC 546	BC 547	BC 548	Unit
Collector–Emitter Voltage	V_{CEO}	65	45	30	Vdc
Collector–Base Voltage	V_{CBO}	80	50	30	Vdc
Emitter–Base Voltage	V_{EBO}	6.0			Vdc
Collector Current — Continuous	I_C	100			mA _{dc}
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625 5.0			mW mW/ $^\circ\text{C}$
Total Device Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	1.5 12			Watt mW/ $^\circ\text{C}$
Operating and Storage Junction Temperature Range	T_J, T_{stg}	–55 to +150			$^\circ\text{C}$

1
2 3

CASE 29–04, STYLE 17
TO–92 (TO–226AA)

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	$^\circ\text{C/W}$
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	$^\circ\text{C/W}$

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
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OFF CHARACTERISTICS

Collector–Emitter Breakdown Voltage ($I_C = 1.0\text{ mA}$, $I_B = 0$)	BC546 BC547 BC548	$V_{(BR)CEO}$	65 45 30	— — —	— — —	V
Collector–Base Breakdown Voltage ($I_C = 100\text{ }\mu\text{A}$)	BC546 BC547 BC548	$V_{(BR)CBO}$	80 50 30	— — —	— — —	V
Emitter–Base Breakdown Voltage ($I_E = 10\text{ }\mu\text{A}$, $I_C = 0$)	BC546 BC547 BC548	$V_{(BR)EBO}$	6.0 6.0 6.0	— — —	— — —	V
Collector Cutoff Current ($V_{CE} = 70\text{ V}$, $V_{BE} = 0$) ($V_{CE} = 50\text{ V}$, $V_{BE} = 0$) ($V_{CE} = 35\text{ V}$, $V_{BE} = 0$) ($V_{CE} = 30\text{ V}$, $T_A = 125^\circ\text{C}$)	BC546 BC547 BC548 BC546/547/548	I_{CES}	— — — —	0.2 0.2 0.2 —	15 15 15 4.0	nA μA

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted) (Continued)

Characteristic	Symbol	Min	Typ	Max	Unit
ON CHARACTERISTICS					
DC Current Gain ($I_C = 10\ \mu\text{A}$, $V_{CE} = 5.0\ \text{V}$)	h_{FE}	—	90	—	—
BC547A/548A		—	150	—	
BC546B/547B/548B		—	270	—	
BC548C					
($I_C = 2.0\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$)		110	—	450	
BC546		110	—	800	
BC547		110	—	800	
BC548		110	180	220	
BC547A/548A		200	290	450	
BC546B/547B/548B		420	520	800	
BC547C/BC548C					
($I_C = 100\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$)		—	120	—	
BC547A/548A		—	180	—	
BC546B/547B/548B		—	300	—	
BC548C					
Collector–Emitter Saturation Voltage ($I_C = 10\ \text{mA}$, $I_B = 0.5\ \text{mA}$)	$V_{CE(sat)}$	—	0.09	0.25	V
($I_C = 100\ \text{mA}$, $I_B = 5.0\ \text{mA}$)		—	0.2	0.6	
($I_C = 10\ \text{mA}$, $I_B = \text{See Note 1}$)		—	0.3	0.6	
Base–Emitter Saturation Voltage ($I_C = 10\ \text{mA}$, $I_B = 0.5\ \text{mA}$)	$V_{BE(sat)}$	—	0.7	—	V
Base–Emitter On Voltage ($I_C = 2.0\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$)	$V_{BE(on)}$	0.55	—	0.7	V
($I_C = 10\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$)		—	—	0.77	
SMALL–SIGNAL CHARACTERISTICS					
Current–Gain — Bandwidth Product ($I_C = 10\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$, $f = 100\ \text{MHz}$)	f_T	150	300	—	MHz
BC546		150	300	—	
BC547		150	300	—	
BC548					
Output Capacitance ($V_{CB} = 10\ \text{V}$, $I_C = 0$, $f = 1.0\ \text{MHz}$)	C_{obo}	—	1.7	4.5	pF
Input Capacitance ($V_{EB} = 0.5\ \text{V}$, $I_C = 0$, $f = 1.0\ \text{MHz}$)	C_{ibo}	—	10	—	pF
Small–Signal Current Gain ($I_C = 2.0\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$, $f = 1.0\ \text{kHz}$)	h_{fe}	125	—	500	—
BC546		125	—	900	
BC547/548		125	220	260	
BC547A/548A		240	330	500	
BC546B/547B/548B		450	600	900	
BC547C/548C					
Noise Figure ($I_C = 0.2\ \text{mA}$, $V_{CE} = 5.0\ \text{V}$, $R_S = 2\ \text{k}\Omega$, $f = 1.0\ \text{kHz}$, $\Delta f = 200\ \text{Hz}$)	NF	—	2.0	10	dB
BC546		—	2.0	10	
BC547		—	2.0	10	
BC548		—	2.0	10	

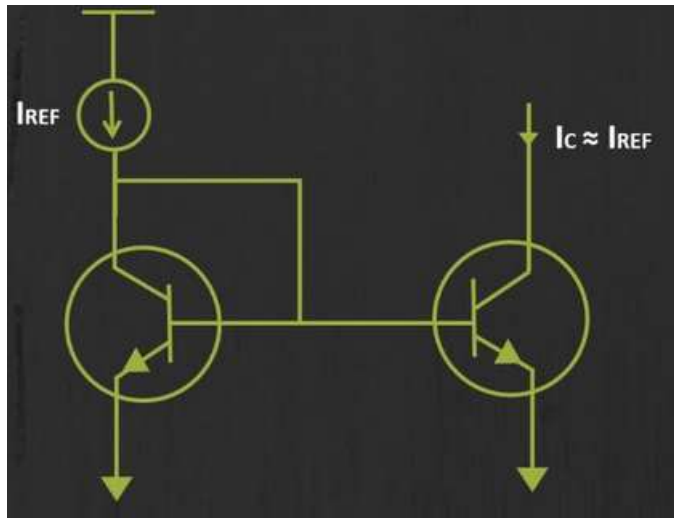
Stages in an Audio Amplifier

1. Current Source
2. Differential Amplifier
3. Subtractor
4. Active Band-pass Filter (BPF)
5. Darlington Amplifier

Also, buffers are added between each stage, except between current source and differential amplifier.

Current Source

- ▶ A current source is the first stage in the audio amplifier and is used to supply current to the differential amplifier.
- ▶ A current source is created using a current mirror made using 2 BJTs.



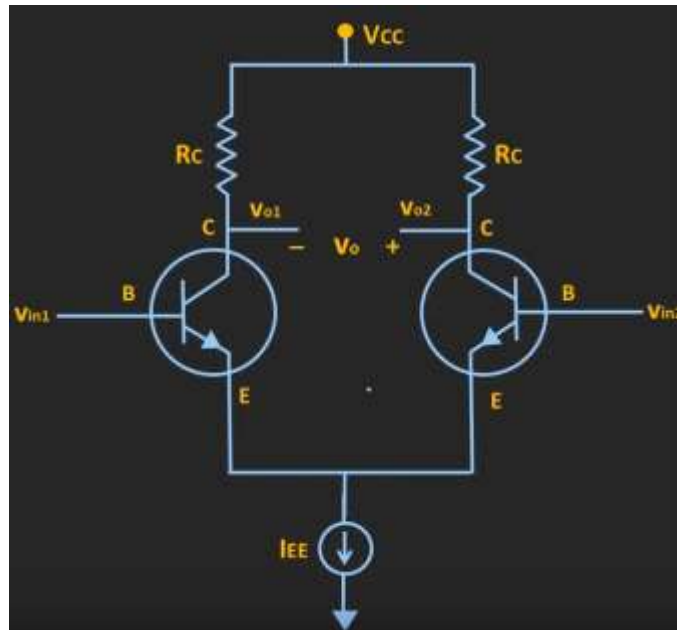
$$I_{C1} \approx I_{S1} \exp \left(\frac{V_{BE}}{V_T} \right)$$

$$I_{REF} \approx I_{S(REF)} \exp \left(\frac{V_{BE}}{V_T} \right)$$

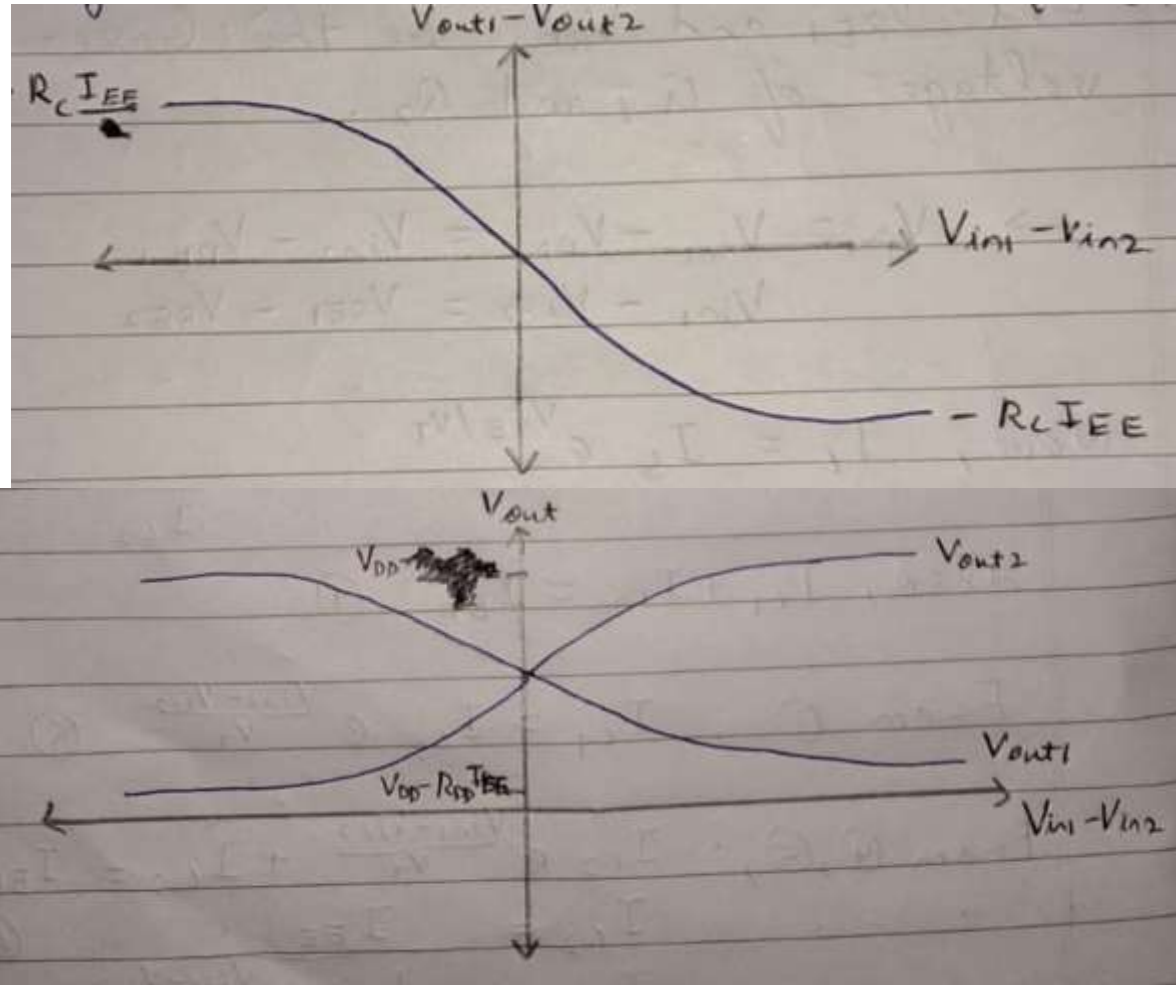
$$\frac{I_{C1}}{I_{REF}} = \frac{I_{S1}}{I_S}$$

Differential Amplifier

- ▶ The differential amplifier is a circuit consisting of 2 identical BJTs or MOSFETs which has 2 inputs and 2 outputs.
- ▶ It amplifies the voltage across the 2 input terminals and gives the amplified output across the 2 output terminals.



Characteristics of Differential Amplifier



Derivation of Differential Amplifier

Let V_{RE1} and V_{RE2} be the base-emitter voltage of Q_1 and Q_2

$$\rightarrow V_p = V_{in1} - V_{RE1} = V_{in2} - V_{RE2}$$

$$V_{in1} - V_{in2} = V_{RE1} - V_{RE2} \quad (1)$$

Now, $I_c = I_s e^{V_{BE}/V_T}$

$$\frac{I_{c1}}{I_{c2}} = e^{V_{RE1}/V_T}$$

$$V_{RE} = V_T \ln \frac{I_c}{I_s} \quad (2)$$

From (1), (2), $V_{in1} - V_{in2} = V_T \ln \frac{I_{c1}}{I_s} - V_T \ln \frac{I_{c2}}{I_s}$

$$V_{in1} - V_{in2} = V_T \ln \frac{I_{c1}}{I_{c2}} \quad (3)$$

Also, $I_{c1} + I_{c2} = I_{EE} \quad (4)$

From (3), $I_{c1} = I_{c2} e^{\frac{V_{in1} - V_{in2}}{V_T}} \quad (5)$

From (4), (5), $I_{c2} e^{\frac{V_{in1} - V_{in2}}{V_T}} + I_{c2} = I_{EE}$

$$I_{c2} = \frac{I_{EE}}{1 + e^{\frac{V_{in1} - V_{in2}}{V_T}}} \quad (6)$$

From (4), (6), $I_{c1} = \frac{I_{EE}}{1 + e^{\frac{V_{in2} - V_{in1}}{V_T}}}$

Now, $V_{out1} = V_{CC} - I_{c1} R_c$

$$V_{out1} = V_{CC} - R_c \frac{I_{EE}}{1 + e^{\frac{V_{in2} - V_{in1}}{V_T}}} \quad (7)$$

Also, $V_{out2} = V_{CC} - I_{c2} R_c$

$$= V_{CC} - R_c \frac{I_{EE}}{1 + e^{\frac{V_{in1} - V_{in2}}{V_T}}} \quad (8)$$

Doing (7) - (8), $V_{out1} - V_{out2} = -R_c \tanh\left(\frac{V_{in1} - V_{in2}}{2V_T}\right)$

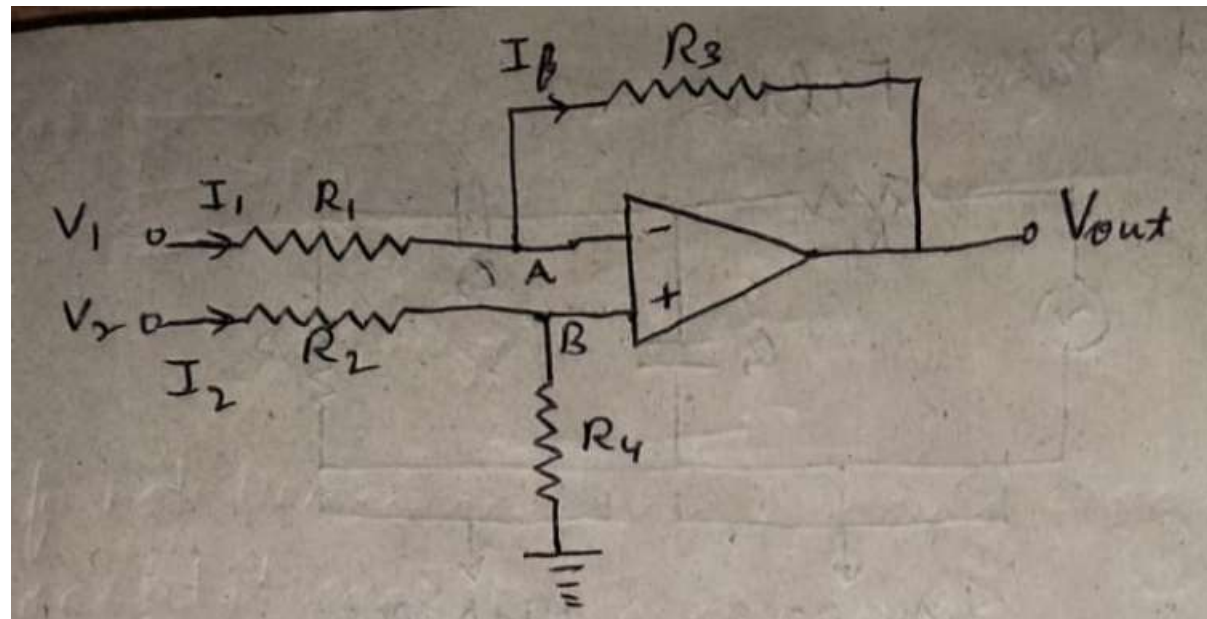
If $\frac{V_{in1} - V_{in2}}{2V_T} \ll 1$, $\tanh\left(\frac{V_{in1} - V_{in2}}{2V_T}\right) \approx \frac{V_{in1} - V_{in2}}{2V_T}$

$$\rightarrow V_{out1} - V_{out2} = -\frac{R_c}{2V_T} (V_{in1} - V_{in2}) I_{EE}$$

$$\frac{V_{out1} - V_{out2}}{V_{in1} - V_{in2}} = -\frac{R_c}{2V_T} I_{EE}$$

Voltage Subtractor

- ▶ A voltage subtractor is a circuit used to produce the voltage difference between 2 terminals.
- ▶ It is made using an op - amp, in which we give 2 input voltages and we get the subtracted voltage at the output.



Derivation of Voltage Subtractor

$$I_1 = \frac{V_1 - V_A}{R_1}, \quad I_2 = \frac{V_2 - V_B}{R_2}, \quad I_3 = \frac{V_A - V_{out}}{R_3}$$

We know, $V_A = V_B$ and $V_B = V_2 \left(\frac{R_4}{R_2 + R_4} \right)$

If $V_2 = 0$, $V_{out(A)} = -V_1 \left(\frac{R_3}{R_1} \right)$

If $V_1 = 0$, $V_{out(B)} = V_2 \left(\frac{R_4}{R_2 + R_4} \right) \left(\frac{R_1 + R_3}{R_1} \right)$

$$\rightarrow V_{out} = -V_{out(A)} + V_{out(B)}$$
$$V_{out} = -V_1 \left(\frac{R_3}{R_1} \right) + V_2 \left(\frac{R_4}{R_2 + R_4} \right) \left(\frac{R_1 + R_3}{R_1} \right)$$

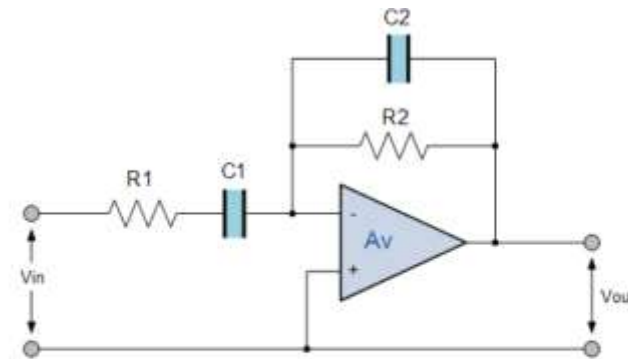
When $R_1 = R_2$ and $R_3 = R_4$,

$$V_{out} = \frac{R_3}{R_1} (V_2 - V_1)$$

When $R_1 = R_2 = R_3 = R_4$, $V_{out} = V_2 - V_1$

Active Band - Pass Filter (BPF)

- ▶ The range of human speech is from 50 Hz to 20 kHz. Any frequency outside this range of frequency will be noise.
- ▶ Hence, an active BPF is used to remove unwanted noise in our audio amplifier.
- ▶ We have used an active filter instead of a passive filter because active filters can be cascaded without loading and impedance matching issues.

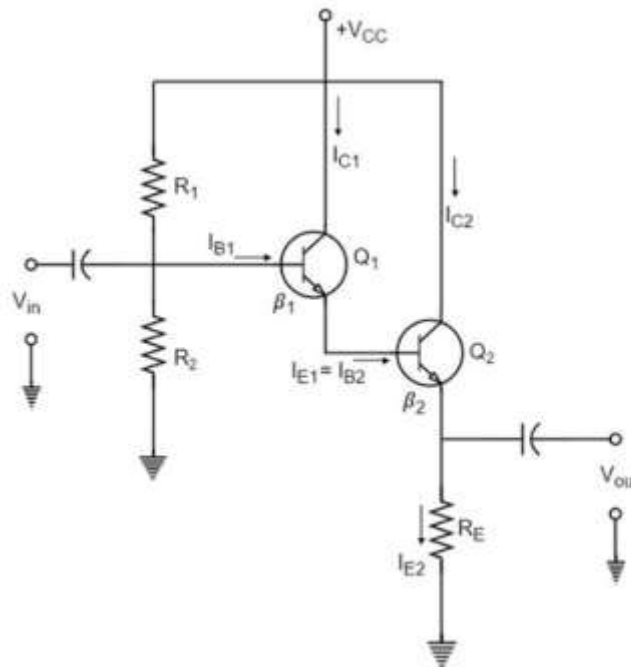


Note: f_{c1} is the lower cut - off frequency and f_{c2} is the upper cut - off frequency

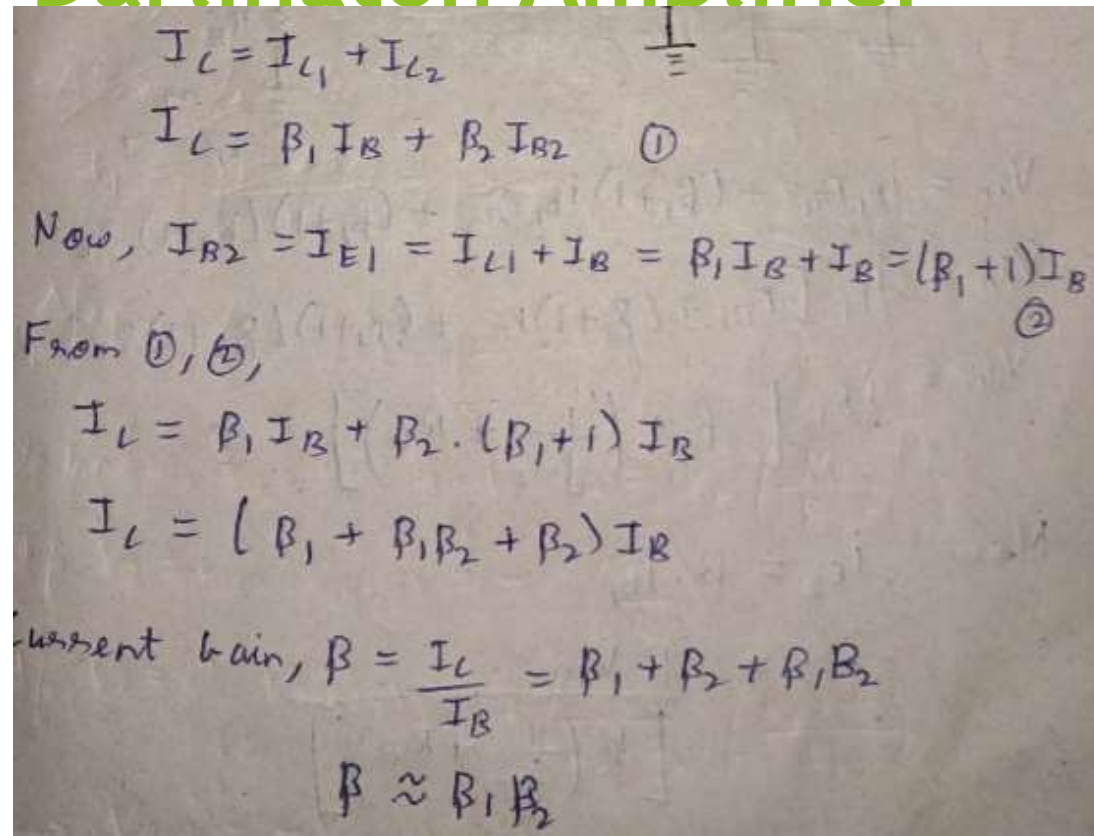
$$\text{Voltage Gain} = -\frac{R_2}{R_1}, \quad f_{c1} = \frac{1}{2\pi R_1 C_1}, \quad f_{c2} = \frac{1}{2\pi R_2 C_2}$$

Darlington Amplifier

- ▶ A Darlington Amplifier is a circuit consisting of 2 BJTs which is used to amplify the input current without much gain in voltage.
- ▶ It is used as the final stage of the audio amplifier to amplify the current to ensure that the speaker gets enough power.



Derivation of Current Gain of Darlington Amplifier



The image shows a piece of paper with handwritten mathematical derivations for the current gain of a Darlington amplifier. The text is written in black ink on a light-colored, slightly textured paper. The derivations are as follows:

$$I_C = I_{C1} + I_{C2}$$
$$I_C = \beta_1 I_B + \beta_2 I_{B2} \quad (1)$$

Now, $I_{B2} = I_{E1} = I_{C1} + I_B = \beta_1 I_B + I_B = (\beta_1 + 1) I_B$ (2)

From (1), (2),

$$I_C = \beta_1 I_B + \beta_2 (\beta_1 + 1) I_B$$
$$I_C = (\beta_1 + \beta_1 \beta_2 + \beta_2) I_B$$

Current gain, $\beta = \frac{I_C}{I_B} = \beta_1 + \beta_2 + \beta_1 \beta_2$

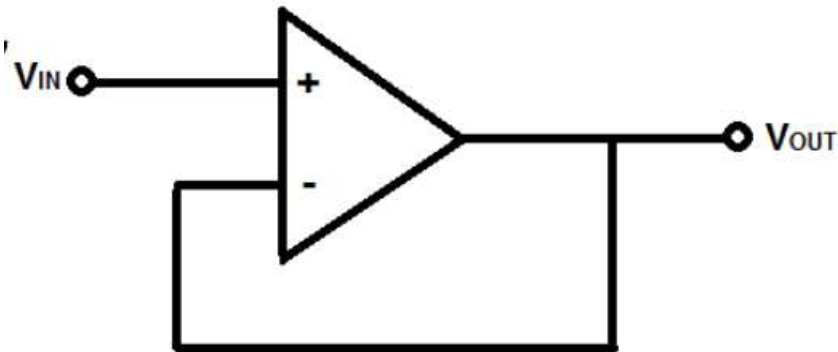
$$\beta \approx \beta_1 \beta_2$$

Small Signal Analysis of Darlington Amplifier

$$\begin{aligned}V_{in} &= i_{b1} r_{\pi1} + (\beta_1 + 1) i_{b1} r_{\pi2} + (\beta_1 + 1) (\beta_2 + 1) R_E \\V_{in} &= i_{b1} [r_{\pi1} + (\beta_1 + 1) r_{\pi2} + (\beta_1 + 1) (\beta_2 + 1) R_E] \\V_{in} &= i_{b1} \left[\beta \times \left(\frac{1}{g_{m2}} + R_E \right) \right] \quad \text{--- (1)} \\ \text{Now, } i_{e2} &= \beta \cdot i_{b1} \\ i_{e2} &= \frac{\beta V_{in}}{\left[\beta \left(\frac{1}{g_{m2}} + R_E \right) \right]} \\ \rightarrow A_v &= \frac{V_{out}}{V_{in}} = \frac{i_{e2} R_E}{V_{in}} \\ A_v &= \frac{R_E}{\frac{1}{g_{m2}} + R_E} \approx 1\end{aligned}$$

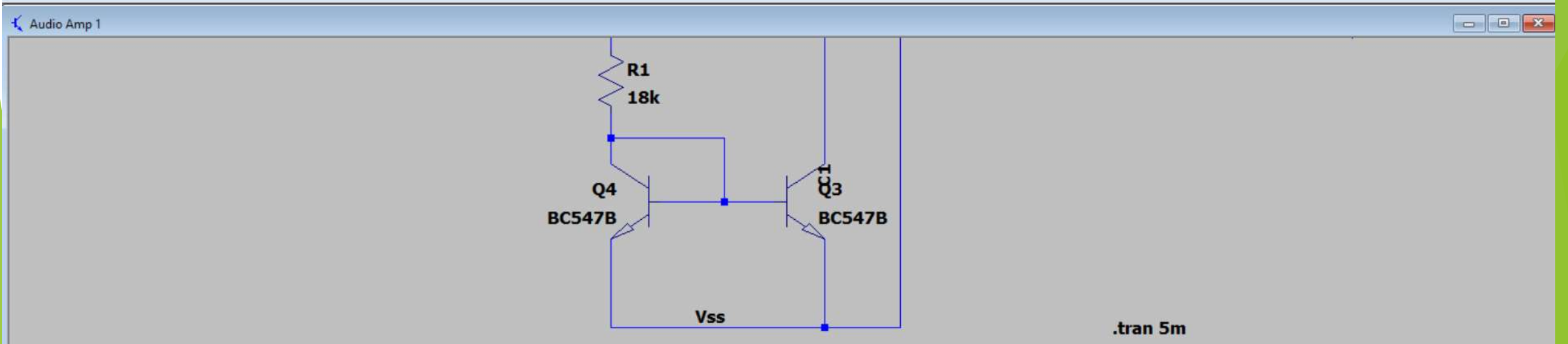
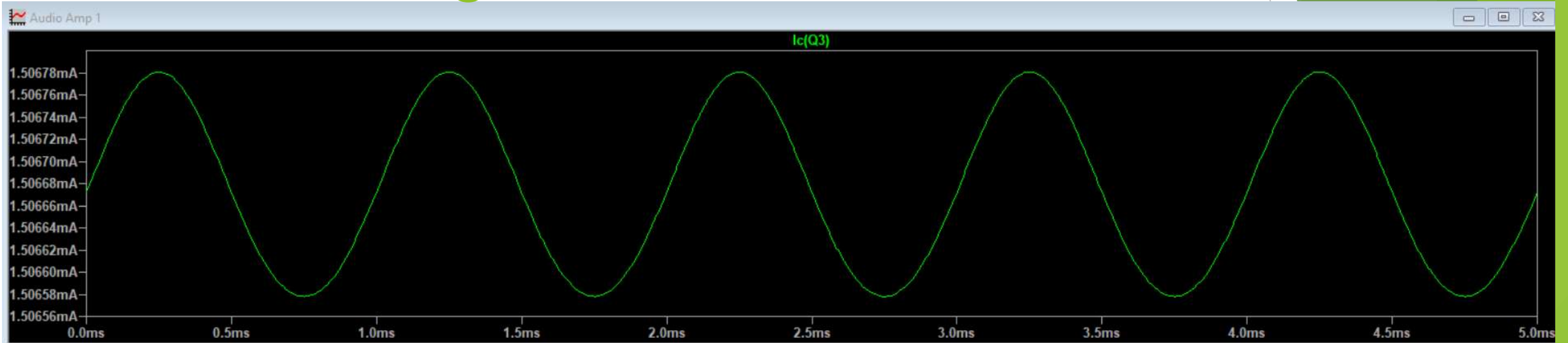
Buffer

- ▶ A unity - gain buffer is a circuit made using a single op - amp which takes an input voltage and gives the same voltage as output without any gain.
- ▶ It is added between each stage have high output impedance for each stage. This prevents loading, interference and impedance matching issues while connecting individual stages with each other.

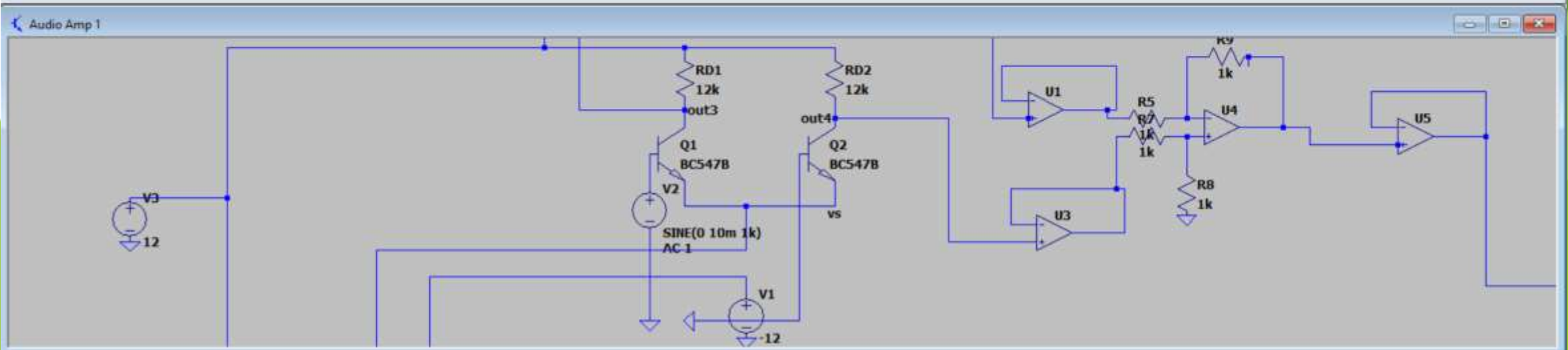
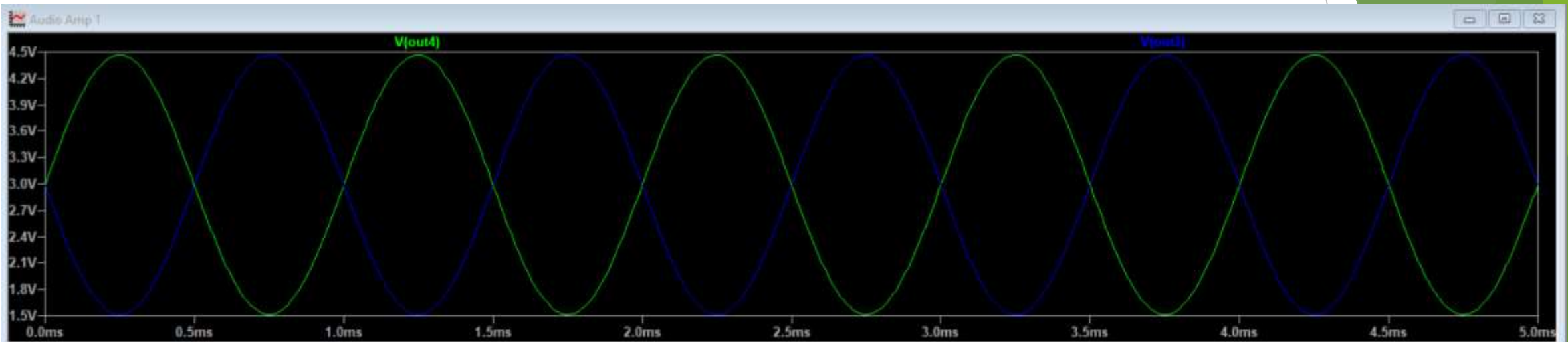


For op-amp, $V_A = V_B$ ①
Also, $V_{in} = V_A$ ②
From ①, ②, $V_{in} = V_B$
 $\rightarrow V_{in} = V_{out}$

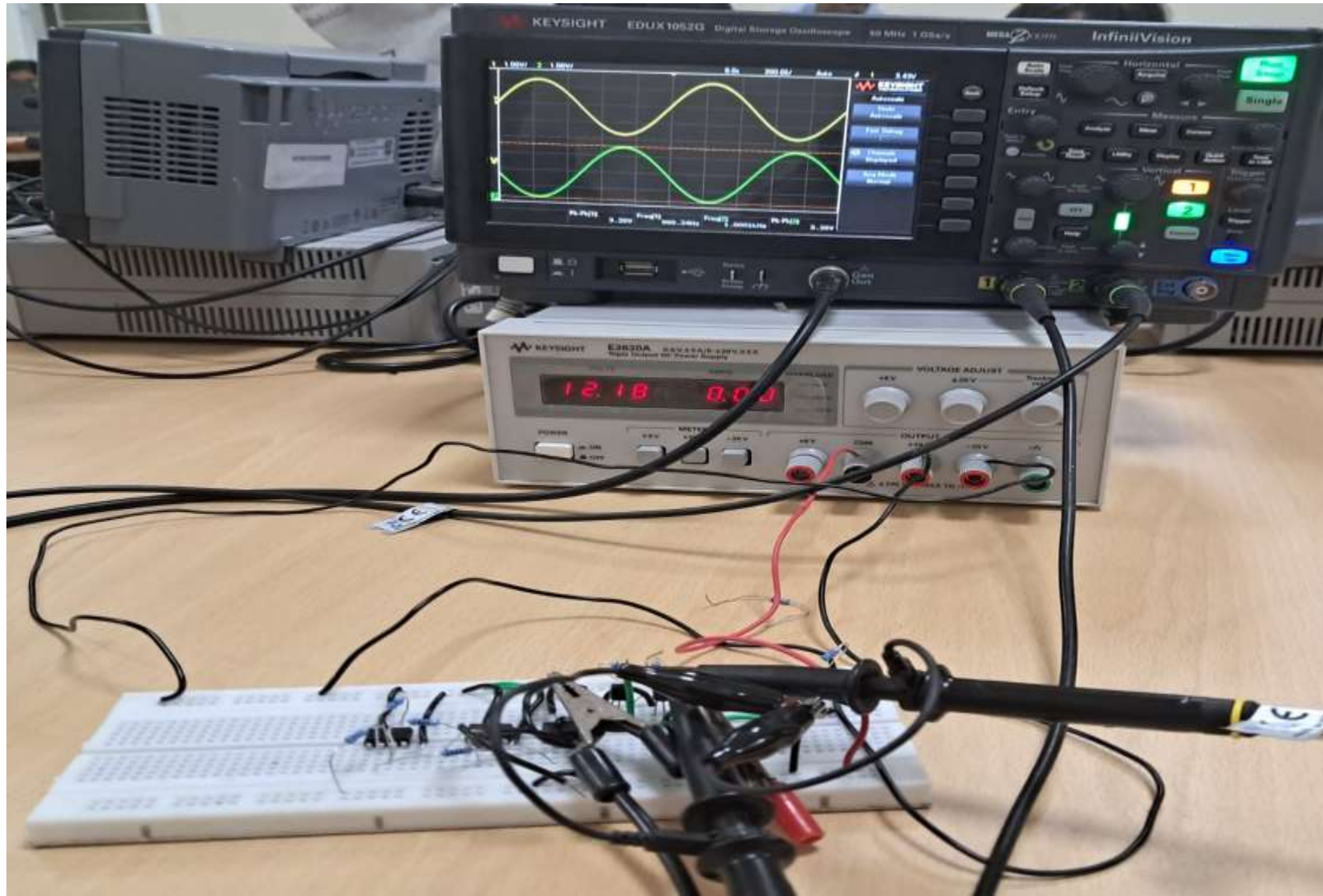
Output of the Current Source made using a Current Mirror



Output at the 2 terminals of the Differential Amplifier



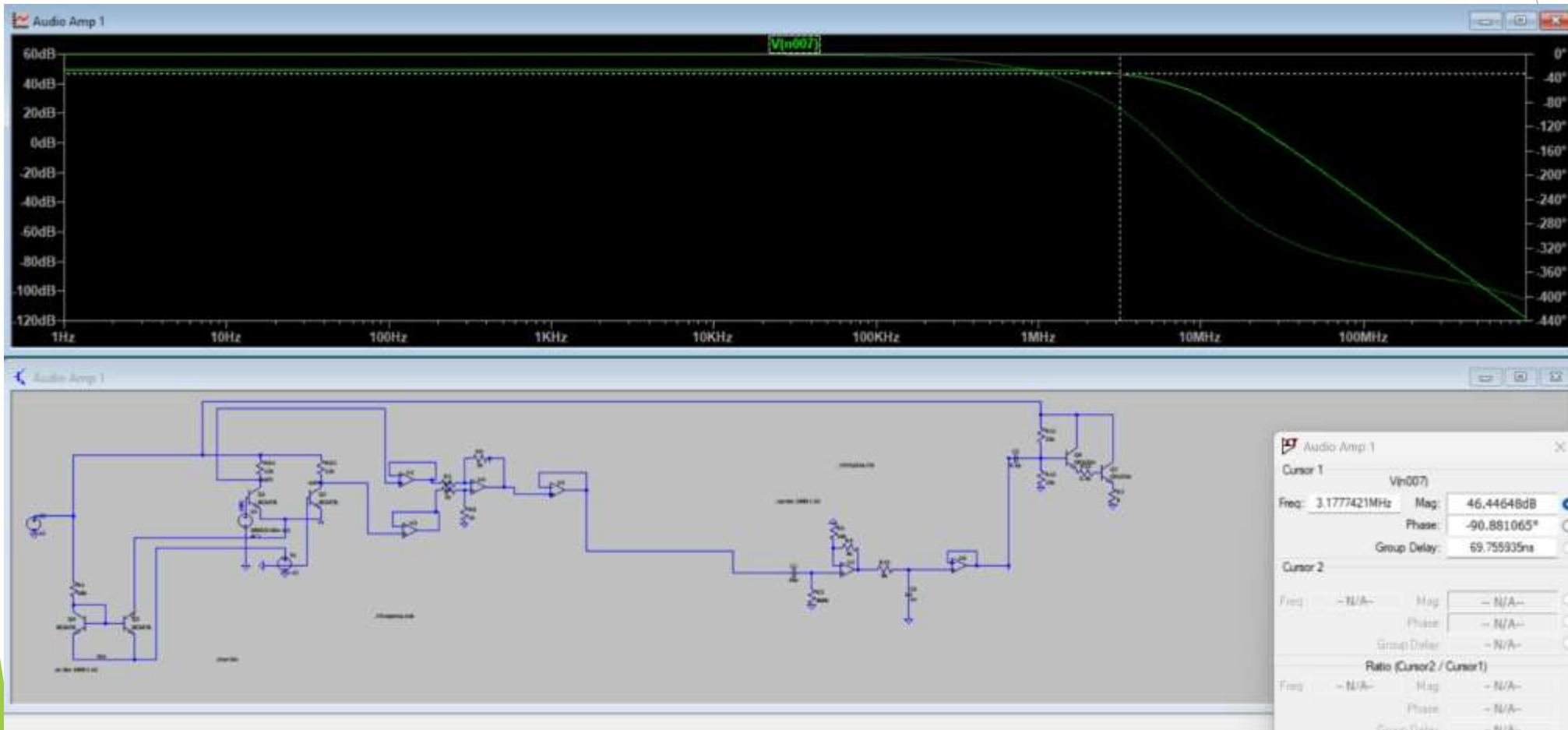
Hardware Output of the Differential Amplifier



Explanation of the above circuit

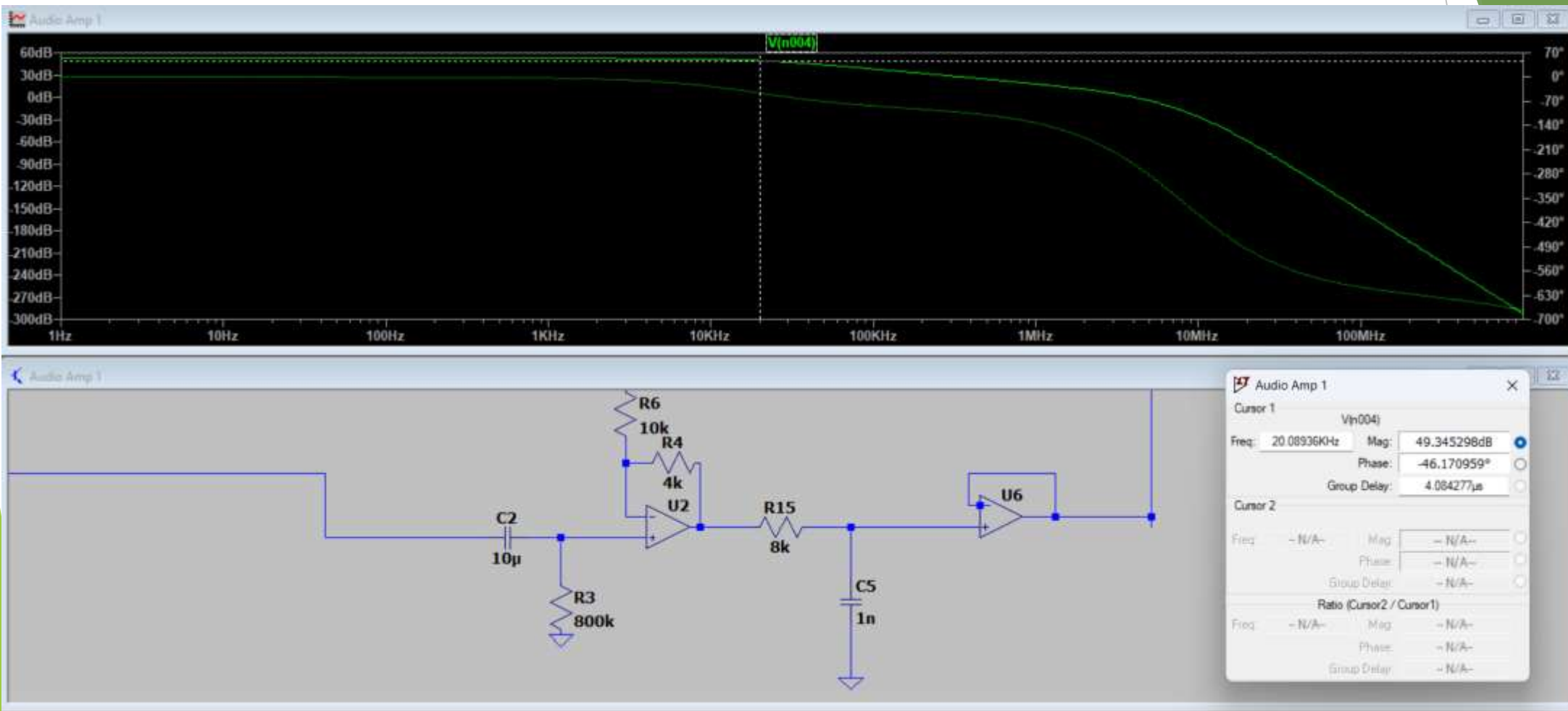
- ▶ A differential amplifier is designed to achieve a gain of 150, i.e. for an input voltage of 20m Vpp, we get an output of approximately 3 Vpp.
- ▶ A voltage subtractor is used to double the obtained gain, as the output measured at the 2 terminals will have a phase shift of exactly 180 degrees.
- ▶ The final output is approximately 6 Vpp for the same input voltage.

Bode Plot before passing the output of the previous stage through a Bandpass Filter



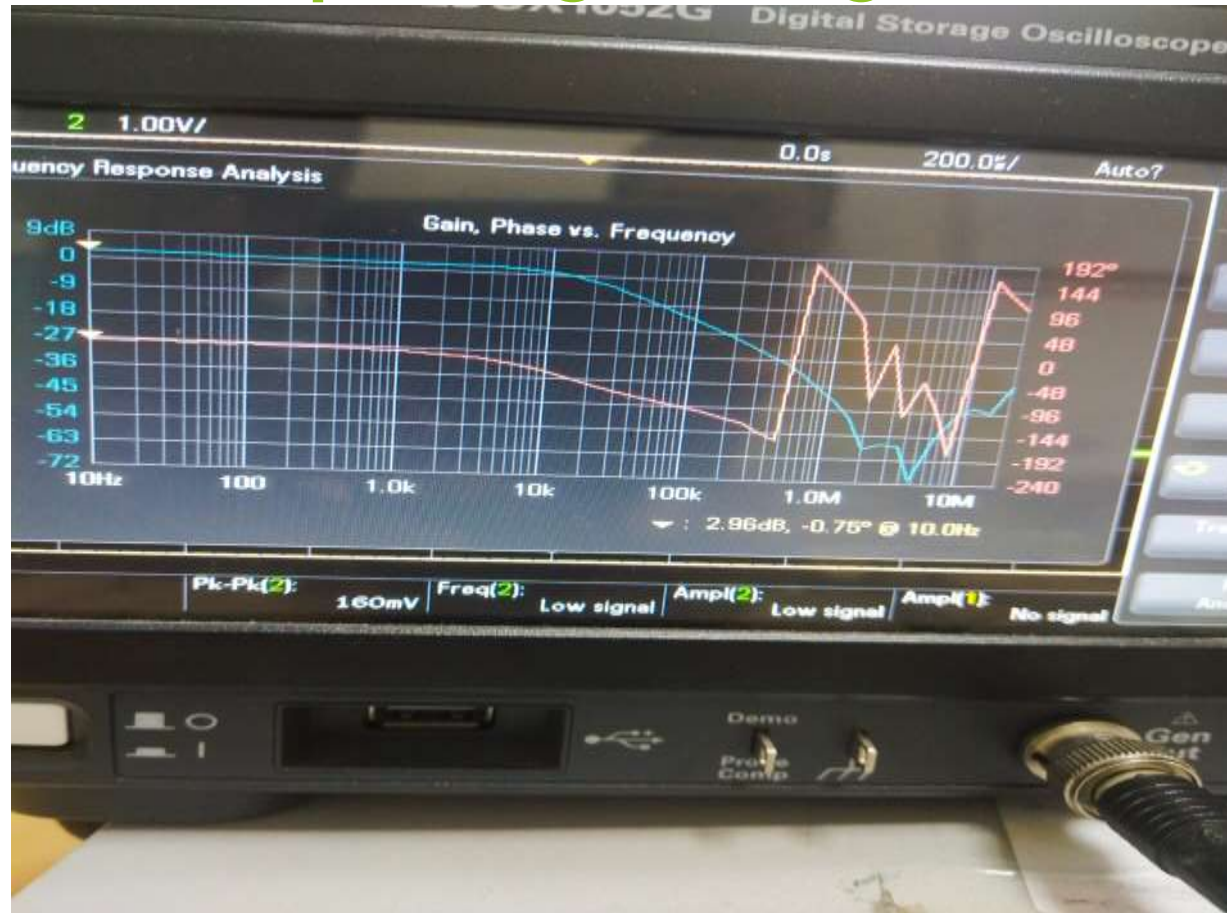
From this bode plot we can see that the -3dB frequency is 3.177 MHz before passing the output of the previous stage through the BPF

Bode Plot after passing the output of the previous stage through a Bandpass Filter



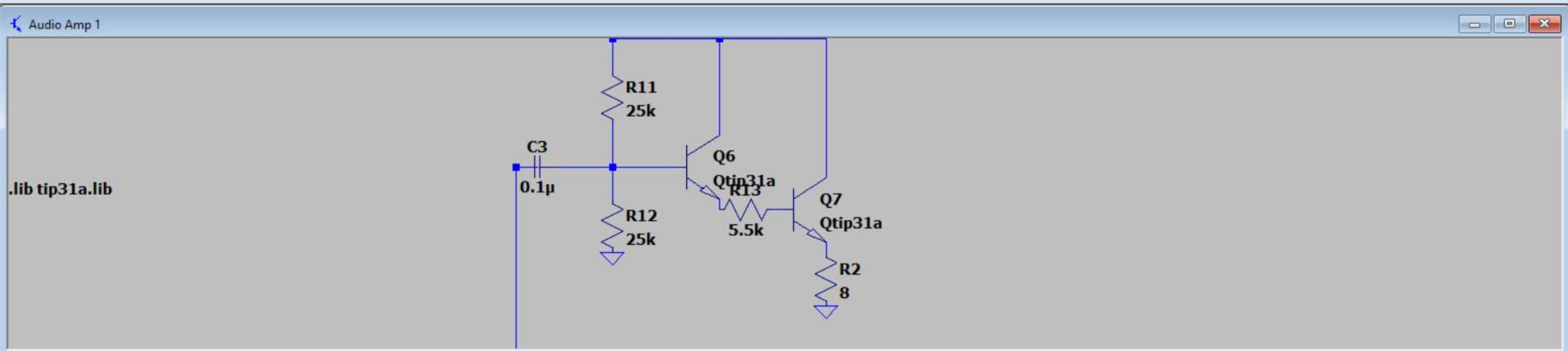
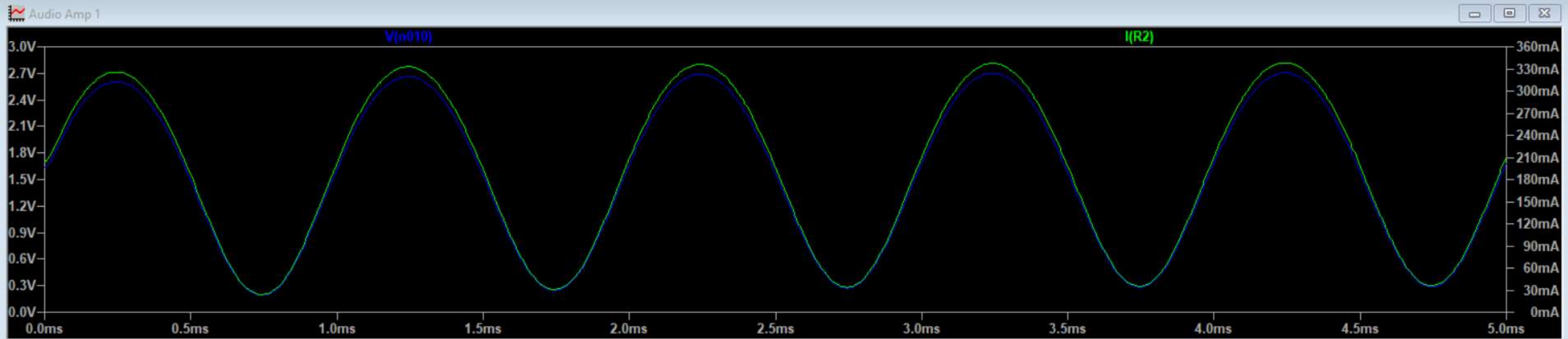
From the bode plot we can see that the -3dB frequency is close to 20 kHz after passing through the BPF, which is what we want.

Bode Plot of the Hardware output after passing through BPF

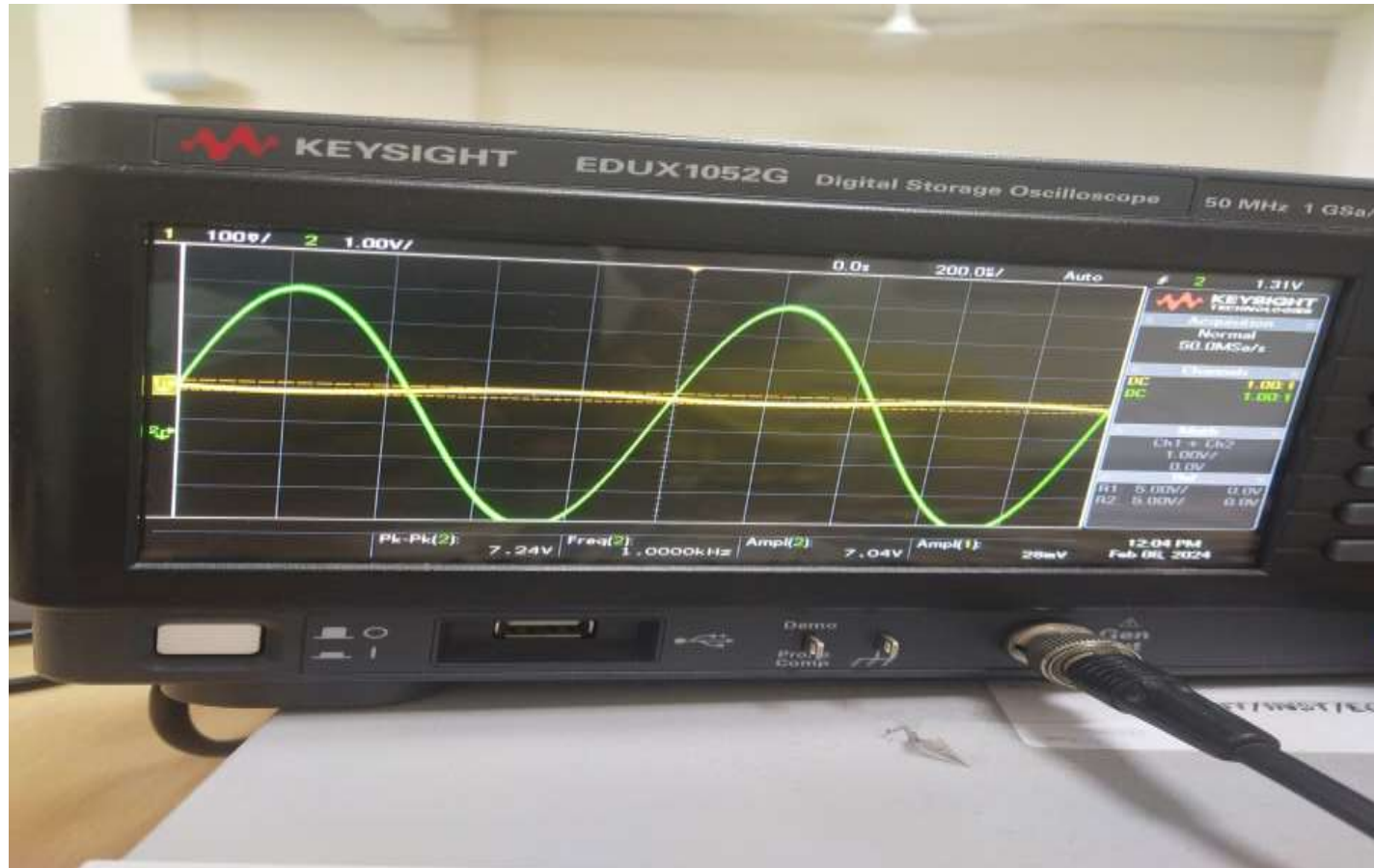


Note: The blue plot represents magnitude and the red line represents the phase. We can notice the -3dB cutoff frequency is very close to 20 kHz which is what we require.

Voltage and Current Output of the Power Amplifier



Final Hardware Output



The above image shows the final output of the hardware audio amplifier. We can see that for a 20m Vpp input we get an output of 7.24 Vpp which is a gain of close to 360.

Power Consumed after Each Stage of Simulation

Stage	Average Voltage (in V)	Average Current (in A)	Average Power (in W)
Differential Amplifier	3	~ 0	~ 0
Subtractor	6	~ 0	~ 0
Bandpass Filter	~ 0	~ 0	~ 0
Power Amplifier	1.7	210m	0.357

Power Consumed after Each Stage of Hardware

Stage	Average Voltage (in V)	Average Current (in A)	Average Power (in W)
Differential Amplifier	3.2	~ 0	~ 0
Subtractor	6.4	~ 0	~ 0
Bandpass Filter	7.20	100u	7.2m
Power Amplifier	7.24	74m	0.535

Observations

1. The simulated and the practically obtained output values are very close to each other if the same components are used in hardware and software.
2. The audio amplifier works for a range of input amplitudes - from 0 to 300 mV.
3. The audio amplifier works perfectly in the desired frequency range of 50 - 20 kHz. We observe significant attenuation outside this frequency range.
4. We get an amplification of close to 360 from our audio amplifier.

Learnings

1. Using buffers between each stage is very important in order to avoid loading and impedance issues.
2. For power amplifiers (TIP31 and TIP32), it is essential to use a heat sink to avoiding heating of the IC. Heating of IC might lead to distortion and degradation in its working.
3. It is better to use an active BPF instead of a passive BPF.
4. For the speaker, both the voltage and current need to be sufficient. If either of them is not enough then the speaker would not work.
5. Sufficient biasing of the microphone is required for it to work.
6. It is very important to check if any of the IC is heated and to replace it immediately to avoid performance issues.
7. For amplification purposes, it is better to use BJT instead of MOSFET.